MAGIC POSITIVITY FOR PSEUDO-SYMMETRIC POLAR DUALS AND CYCLE POLARS

HIDEFUMI OHSUGI, AKIYOSHI TSUCHIYA

Editorial note. Complete original mathematical paper74–2257, all divided-difference/local-positivity/classification proofs and unchanged bibliography.bib retained. Authors retain responsibility for mathematics; no proof repair, added hypotheses, independent author review or certification.

Editorial note. Both original main polar-family applications are one counted object: source2131–2133 explicitly obtains the cycle conclusion from the same Q_d positivity theorem. Ehrhart/real-rooted/gamma corollaries and the corrective appendix for the prior primal-polytope proof are context only, not additional counts.

Editorial note. Original AI declaration2252–2255 retained exactly: ChatGPT exploratory proof-strategy discussions, English editing and LaTeX assistance; authors state verification of arguments/references. This is author disclosure, not agent certification or AI-free assertion.

Editorial note. Public neighbors165780 and211239 compared by source/target metadata and the current authors stated prior-work scope, without old proof re-audit.

Allowed standard prerequisites (editorial).

- Ehrhart and magic transforms, reflexive polars, pseudo-symmetric classification, divided differences and binary codes. Original pseudo-symmetric simplicial reflexive/smooth-Fano hypotheses, coefficient basis k^i(k+1)^(d-i), cycle-polar dual lattice conventions and all original classification/closure formulas retained. All source-local local decomposition and positivity proofs included, with exact standard external citations.. Full original mathematical paper, both main deductions and original corrective appendix supplied; no independently reconstructed argument or proof certification.

1. Introduction

Let $P \subset \mathbb{R}^d$ be a d-dimensional lattice polytope. Its Ehrhart polynomial is

$$L_P(k) = |kP \cap \mathbb{Z}^d|,$$

and its Ehrhart series is

$$1 + \sum_{k \geq 1} L_P(k) t^k = \frac{h^*(P;t)}{(1-t)^{d+1}}.$$

We refer to [4] for the basic theory of Ehrhart polynomials and Ehrhart series. The numerator $h^*(P;t)$ is the h^*-polynomial of P, whose coefficients are nonnegative integers by Stanley's theorem [24]. If P is Gorenstein, then $h^*(P;t)$ is palindromic. Writing $s = \deg h^*(P;t)$, it therefore has a unique expansion

$$h^*(P;t) = \sum_{i=0}^{\lfloor s/2 \rfloor} \gamma_i t^i (1+t)^{s-2i}.$$

The polynomial $h^*(P;t)$ is called *γ-positive* if $\gamma_i \geq 0$ for every i.

One of the central positivity conjectures in algebraic and geometric combinatorics is Gal's conjecture [14], which predicts γ-positivity of the h-polynomial of every flag simplicial sphere. Motivated by this conjecture, Ferroni and Higashitani proposed an Ehrhart-theoretic analogue. They conjectured that a Gorenstein lattice polytope admitting a quadratic triangulation has a γ-positive h^*-polynomial [13, Conjecture 3.28]. Here a quadratic triangulation means a regular unimodular flag triangulation.

Another long-standing problem concerns the toric ideals of smooth lattice polytopes. The Bøgvad conjecture predicts that the toric ideal of every smooth lattice polytope is generated by quadratic binomials. A stronger version asks for a quadratic Gröbner basis, while a squarefree quadratic initial ideal corresponds to a quadratic triangulation via the standard correspondence between initial ideals of toric

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ideals and regular triangulations; see [25, Chapter 8]. Although these statements are logically distinct, together with the conjecture of Ferroni and Higashitani they suggest the following positivity problem.

Conjecture 1.1. *Every smooth Gorenstein lattice polytope has a γ-positive h^*-polynomial.*

For background and classification results on smooth Gorenstein polytopes, see [19].

The purpose of this paper is to establish a stronger positivity property for the polar duals of pseudo-symmetric simplicial reflexive polytopes. A reflexive polytope is called *pseudo-symmetric* if it has a pair of opposite facets.

Nill classified pseudo-symmetric simplicial reflexive polytopes in arbitrary dimension [21]. His classification decomposes every such polytope into a free sum of a centrally symmetric reflexive crosspolytope, even-dimensional del Pezzo polytopes, and pseudo-del Pezzo polytopes. The smooth Fano members form an important subclass, classified earlier by Ewald [9, 10]. The smooth Fano subclass has been particularly tractable in several aspects of toric geometry. For instance, Haase used these polytopes and their polar duals to compute stringy Hodge numbers of generic hypersurfaces in the associated toric varieties and to study related questions concerning mirror symmetry and crepant resolutions [15]. The toric Fano varieties associated with the centrally symmetric members of this family have also been studied in connection with Tian's α-invariant and Kähler–Einstein metrics [3, 23]. On the Ehrhart side, in our previous work [22, Theorem 5.8], we stated that the h^*-polynomial of every pseudo-symmetric simplicial reflexive polytope is γ-positive. The proof there invokes Nill's decomposition, but the centrally symmetric reflexive crosspolytope factor was inadvertently treated as a standard crosspolytope. We supply the missing argument in Appendix A. The present paper instead studies the polar duals and proves the stronger property of magic positivity. These structural, geometric, and enumerative properties make the polar duals of pseudo-symmetric smooth Fano polytopes a natural testing ground for Conjecture 1.1.

We recall the relevant geometric terminology and the correspondence between smooth Fano polytopes and smooth reflexive polytopes in Section 2. Our first main result is the following.

Theorem 1.2. *Let P be a pseudo-symmetric simplicial reflexive polytope. Then the Ehrhart polynomial $L_{P^\vee}(k)$ is magic positive. If P is a smooth Fano polytope, then all magic coefficients of $L_{P^\vee}(k)$ are strictly positive.*

Here, following Ferroni and Higashitani [13, Section 4.5], a polynomial $L(k)$ of degree d has a unique expansion

$$L(k) = \sum_{i=0}^{d} a_i k^i (k+1)^{d-i}.$$

The numbers $a_0, \ldots, a_d$ are called the *magic coefficients* of L, and L is called *magic positive* if $a_i \geq 0$ for every i. We call

$$M_L(y) := (1-y)^d L\left(\frac{y}{1-y}\right) = \sum_{i=0}^{d} a_i y^i$$

the *magic transform* of L. Thus L is magic positive if and only if $M_L(y)$ has nonnegative coefficients. A lattice polytope P is called *Ehrhart positive* if all coefficients of $L_P(k)$ in the standard monomial basis are positive. If $L(k) = L_P(k)$, then $a_0 = L_P(0) = 1$, so magic positivity of $L_P(k)$ implies Ehrhart positivity of P. It also implies real-rootedness of the corresponding h^*-polynomial by the Pólya-frequency-preserving transformation studied by Brändén; see [5] and [13, Theorem 4.19]. For a reflexive polytope, real-rootedness in turn implies γ-positivity.

Recent work has established magic positivity for other families, including Pitman–Stanley polytopes and broad classes of partial permutohedra [2, 18].

We therefore obtain the following consequences.

Corollary 1.3. *Let P be a pseudo-symmetric simplicial reflexive polytope. Then $P^\vee$ is Ehrhart positive, and $h^*(P^\vee; t)$ is real-rooted and γ-positive.*

In particular, if P is smooth Fano, then $P^\vee$ is a smooth reflexive, and hence smooth Gorenstein, lattice polytope. Thus Theorem 1.2 verifies Conjecture 1.1 for the polar duals of pseudo-symmetric smooth Fano polytopes.

Haase proved that the polar dual of every pseudo-symmetric smooth Fano polytope admits a quadratic triangulation [15, Corollaries IV.2.6 and IV.2.10]. Thus these polytopes satisfy the triangulation hypothesis in the conjecture of Ferroni and Higashitani. On the other hand, Ferroni constructed Gorenstein lattice polytopes admitting quadratic triangulations whose h^*-polynomials are not log-concave

[12, Theorem 1.4 and Section 6]. Since magic positivity implies real-rootedness of the corresponding h^*-polynomial, whose nonnegative coefficients are then log-concave by Newton's inequalities, these examples show that the quadratic-triangulation hypothesis alone does not imply magic positivity, even in the Gorenstein case.

A second application concerns symmetric edge polytopes, introduced and studied in [20]. For a graph G on $[n] := \{1, 2, \ldots, n\}$, its symmetric edge polytope is

$$\mathcal{P}_G = \mathrm{conv}\{\pm(e_i - e_j) : \{i, j\} \in E(G)\},$$

where e_i is the i-th standard unit vector of $\mathbb{R}^n$. For a connected graph G, we regard $\mathcal{P}_G$ as a full-dimensional lattice polytope in $H = \{x \in \mathbb{R}^n : x_1 + \cdots + x_n = 0\}$, equipped with the lattice $H \cap \mathbb{Z}^n$, and take its polar dual with respect to the dual lattice. Konoike studied magic positivity of the polar duals associated with cycles. For the cycle C_n, he obtained an explicit Ehrhart formula [17, Proposition 4.6], proved positivity of the first three and the last three magic coefficients [17, Theorem 1.4], and asked whether all magic coefficients are nonnegative [17, Question 4.5]. We answer this question in a stronger form.

Theorem 1.4. *For every $n \geq 3$, all magic coefficients of $L_{\mathcal{P}_{C_n}^\vee}(k)$ are strictly positive.*

Corollary 1.5. *For every $n \geq 3$, the polytope $\mathcal{P}_{C_n}^\vee$ is Ehrhart positive, and its h^*-polynomial is real-rooted and γ-positive.*

The strict positivity results for the del Pezzo and cycle cases are based on a positivity result for divided differences, while the pseudo-del Pezzo case follows from an averaging identity. The general pseudo-symmetric simplicial reflexive case is then obtained from Nill's classification and the binary-code description of centrally symmetric reflexive crosspolytopes, while Ewald's classification yields strict positivity in the smooth Fano case.

The paper is organized as follows. Section 2 collects the preliminaries and introduces the family of polytopes underlying the proofs. Section 3 recalls the divided-difference identities used in the proof. Section 4 develops the local decomposition of the Ehrhart and magic coefficients. Section 5 establishes the positivity of the local terms. Finally, Section 6 proves the main result for pseudo-symmetric simplicial reflexive polytopes and applies these results to polar duals of symmetric edge polytopes of cycles.

Acknowledgment. This work was supported by JSPS KAKENHI Grant Numbers JP22K13890, JP24K00534, and JP26K00618.

2. Preliminaries and reduction

We refer to [6] for background on lattice polytopes and toric varieties. We first recall the terminology used throughout the paper. Let $M \simeq \mathbb{Z}^d$ be a lattice, let $N = \mathrm{Hom}(M, \mathbb{Z})$ be its dual lattice, and write $M_\mathbb{R} = M \otimes_\mathbb{Z} \mathbb{R}$ and $N_\mathbb{R} = N \otimes_\mathbb{Z} \mathbb{R}$. For a full-dimensional polytope $P \subset M_\mathbb{R}$ containing the origin in its interior, its *polar dual* is

$$P^\vee = \{y \in N_\mathbb{R} : \langle y, x \rangle \leq 1 \text{ for every } x \in P\}.$$

A lattice polytope $P \subset M_\mathbb{R}$ is called *reflexive* if the origin lies in its interior and $P^\vee$ is again a lattice polytope. A full-dimensional lattice polytope containing the origin in its interior is called *pseudo-symmetric* if it has a pair of opposite facets. A lattice polytope P is called *Gorenstein* if $rP - m$ is reflexive for some positive integer r and some lattice point $m \in M$. The integer r is called the *index* of P. A lattice polytope is Gorenstein if and only if its h^*-polynomial is palindromic [16, 13].

A d-dimensional lattice polytope $P \subset M_\mathbb{R}$ is called *smooth* if, at every vertex v of P, the primitive lattice vectors along the d edges emanating from v form a basis of M. A full-dimensional lattice polytope $R \subset N_\mathbb{R}$ containing the origin in its interior is called a *Fano polytope* if all its vertices are primitive lattice points. A Fano polytope is called *smooth Fano* if the vertices of every facet form a basis of N. We use the conventions of [9, 10, 15].

A lattice polytope is smooth if and only if the projective toric variety defined by its normal fan is smooth [6, Theorem 2.4.3]. Reflexive polytopes are the anticanonical polytopes of Gorenstein toric Fano varieties [6, Theorem 8.3.4].

Polar duality identifies smooth Fano polytopes with smooth reflexive polytopes. Indeed, if the vertices $v_1, \ldots, v_d$ of a facet of R form a lattice basis, then the corresponding vertex of $R^\vee$ is $v_1^* + \cdots + v_d^*$ and its primitive edge directions are $-v_1^*, \ldots, -v_d^*$, where $v_1^*, \ldots, v_d^*$ is the dual basis. The converse follows by considering the primitive facet normals of a smooth reflexive polytope.

For full-dimensional polytopes $P \subset \mathbb{R}^d$ and $Q \subset \mathbb{R}^e$ containing the origin in their interiors, define their *free sum* by

$$P \oplus Q = \operatorname{conv}\big((P \times \{0\}) \cup (\{0\} \times Q)\big).$$

We will use the following elementary properties.

Lemma 2.1. *Let P and Q be lattice polytopes.*

(1) *If P and Q contain the origin in their interiors, then*

$$(P \oplus Q)^\vee = P^\vee \times Q^\vee.$$

(2) *Their Ehrhart polynomials satisfy*

$$L_{P\times Q}(k) = L_P(k) L_Q(k).$$

(3) *If $L_P(k)$ and $L_Q(k)$ are magic positive, then so is $L_{P\times Q}(k)$. If all magic coefficients of both factors are strictly positive, then the same holds for their product.*

For $m \geq 1$, let

$$V_{2m} = \operatorname{conv}\left(\pm e_1, \ldots, \pm e_{2m}, \pm(e_1 + \cdots + e_{2m})\right)$$

be the del Pezzo polytope, and let

$$\widetilde{V}_{2m} = \operatorname{conv}\left(\pm e_1, \ldots, \pm e_{2m}, -(e_1 + \cdots + e_{2m})\right)$$

be the pseudo-del Pezzo polytope.

We use Nill's classification of pseudo-symmetric simplicial reflexive polytopes [21, Theorem 0.1].

Proposition 2.2 (Nill)**.** *Every pseudo-symmetric simplicial reflexive polytope is lattice equivalent to a free sum*

$$R \oplus V_{2m_1} \oplus \cdots \oplus V_{2m_r} \oplus \widetilde{V}_{2n_1} \oplus \cdots \oplus \widetilde{V}_{2n_s},$$

where $r, s \geq 0$, all $m_i, n_j \geq 1$, and R is a centrally symmetric reflexive crosspolytope. The factor R may be omitted.

For the smooth Fano subclass, Ewald's classification gives the following more restrictive decomposition [9, 10]; see also [15, Theorem IV.2.1].

Proposition 2.3 (Ewald)**.** *Every pseudo-symmetric smooth Fano polytope is lattice equivalent to a free sum of copies of $[-1, 1]$, del Pezzo polytopes, and pseudo-del Pezzo polytopes.*

The interval $[-1, 1]$ is self-dual and has Ehrhart polynomial

$$2k + 1 = k + (k + 1),$$

which is magic positive.

2.1. **Centrally symmetric reflexive crosspolytopes.** Let

$$R = \operatorname{conv}(\pm v_1, \ldots, \pm v_d)$$

be a d-dimensional centrally symmetric reflexive crosspolytope in a lattice N. Since $v_1, \ldots, v_d$ form an $\mathbb{R}$-basis of $N_\mathbb{R}$, let

$$\varphi : N_\mathbb{R} \longrightarrow \mathbb{R}^d$$

be the linear isomorphism defined by $\varphi(v_i) = e_i$, and put

$$\Lambda := \varphi(N).$$

Then

$$\mathbb{Z}^d \subseteq \Lambda \subseteq \mathbb{R}^d, \qquad R = \operatorname{conv}(\pm e_1, \ldots, \pm e_d).$$

Thus

$$R^\vee = [-1, 1]^d$$

with respect to the dual lattice $\Lambda^\vee$.

Since R is reflexive, every vertex of $R^\vee$, that is, every sign vector $\varepsilon \in \{\pm 1\}^d$, belongs to $\Lambda^\vee$. Hence, for every $x = (x_1, \ldots, x_d) \in \Lambda$,

$$\varepsilon_1 x_1 + \cdots + \varepsilon_d x_d \in \mathbb{Z} \qquad (\varepsilon \in \{\pm 1\}^d).$$

Comparing two sign vectors which differ only in the i-th coordinate gives

$$2x_i \in \mathbb{Z} \qquad (1 \leq i \leq d),$$

while taking $\varepsilon = (1, \dots, 1)$ gives

$$x_1 + \cdots + x_d \in \mathbb{Z}.$$

In particular,

$$\mathbb{Z}^d \subseteq \Lambda \subseteq \frac{1}{2}\mathbb{Z}^d.$$

Define

$$\pi : \Lambda \longrightarrow \mathbb{F}_2^d, \qquad \pi(x) = (2x_1, \dots, 2x_d) \pmod 2.$$

Its kernel is $\mathbb{Z}^d$, and hence

$$C := \pi(\Lambda) \cong \Lambda/\mathbb{Z}^d$$

is a binary linear code. For $c \in \mathbb{F}_2^d$, let $\mathrm{wt}(c)$ denote its Hamming weight. Since $x_1 + \cdots + x_d \in \mathbb{Z}$ for every $x \in \Lambda$, every codeword of C has even Hamming weight. Thus C is an even binary linear code.

Let

$$D := C^\perp = \left\{ b \in \mathbb{F}_2^d : \sum_{j=1}^{d} b_j c_j = 0 \text{ in } \mathbb{F}_2 \text{ for every } c \in C \right\}$$

be the dual code of C. For $0 \le i \le d$, let

$$A_i^\perp = \#\{b \in D : \mathrm{wt}(b) = i\}.$$

Proposition 2.4. *With the notation above,*

$$L_{R^\vee}(k) = \sum_{i=0}^{d} A_i^\perp k^i (k+1)^{d-i}.$$

In particular, $L_{R^\vee}(k)$ is magic positive.

Proof. Since $\mathbb{Z}^d \subseteq \Lambda$, we have

$$\Lambda^\vee \subseteq \mathbb{Z}^d.$$

We first describe which elements of $\mathbb{Z}^d$ belong to $\Lambda^\vee$.

Let $z \in \mathbb{Z}^d$. For each $x \in \Lambda$, let $c \in C \subseteq \mathbb{F}_2^d$ be the image of x under the map

$$\pi : \Lambda \longrightarrow \mathbb{F}_2^d, \qquad \pi(x) = (2x_1, \dots, 2x_d) \pmod 2,$$

and identify c with its representative in $\{0,1\}^d$. Then

$$x = n + \frac{1}{2}c$$

for some $n \in \mathbb{Z}^d$. Hence

$$\langle z, x \rangle = z \cdot n + \frac{1}{2} z \cdot c.$$

Since $z \cdot n \in \mathbb{Z}$, the condition $\langle z, x \rangle \in \mathbb{Z}$ for every $x \in \Lambda$ is equivalent to

$$z \cdot c \equiv 0 \pmod 2 \qquad \text{for every } c \in C.$$

Equivalently,

$$(z \bmod 2) \cdot c = 0 \qquad \text{in } \mathbb{F}_2$$

for every $c \in C$. Thus

$$\Lambda^\vee = \{z \in \mathbb{Z}^d : z \bmod 2 \in C^\perp\} = \{z \in \mathbb{Z}^d : z \bmod 2 \in D\}.$$

Now let k be a nonnegative even integer. Since

$$R^\vee = [-1, 1]^d,$$

we have

$$kR^\vee \cap \Lambda^\vee = \{z \in [-k, k]^d \cap \mathbb{Z}^d : z \bmod 2 \in D\}.$$

The interval $[-k, k] \cap \mathbb{Z}$ contains $k+1$ even integers and k odd integers. Therefore, for a fixed $b \in D$ with $\mathrm{wt}(b) = i$, the number of $z \in [-k, k]^d \cap \mathbb{Z}^d$ satisfying

$$z \bmod 2 = b$$

is

$$k^i (k+1)^{d-i}.$$

Summing over all elements of D, we obtain

$$L_{R^\vee}(k) = \sum_{i=0}^{d} A_i^{\perp} k^i (k+1)^{d-i}$$

for every nonnegative even integer k.

Both sides are polynomials in k, and they agree for infinitely many values of k. Hence the identity holds for all k. Since $A_i^{\perp} \geq 0$ for every i, these are precisely the magic coefficients of $L_{R^\vee}(k)$, and therefore $L_{R^\vee}(k)$ is magic positive. □

2.2. **The slab polytopes Q_d.** For $d \geq 2$, put

$$Q_d = \left\{ x \in [-1,1]^d : -1 \leq x_1 + \cdots + x_d \leq 1 \right\}.$$

The vertex description of V_{2m} immediately gives

$$Q_{2m} = V_{2m}^{\vee}.$$

Konoike showed that Q_{n-1} is lattice equivalent to $\mathcal{P}_{C_n}^{\vee}$ and obtained the following Ehrhart formula [17, Proposition 4.6 and its proof]:

$$L_{Q_d}(k) = \sum_{j=0}^{\lfloor d/2 \rfloor} (-1)^j \binom{d+1}{j} \binom{(d+1-2j)k + d - j}{d}. \tag{1}$$

For even d, the same formula is also contained in Haase's explicit Ehrhart formulas [15, Proposition IV.2.7].

We now rewrite (1) in a form adapted to magic positivity. Let

$$d = 2m + \varepsilon, \qquad \varepsilon \in \{0,1\}, \qquad \alpha = \frac{1+\varepsilon}{2},$$

and set

$$\lambda_q = q + \alpha \qquad (0 \leq q \leq m).$$

Proposition 2.5. *There is a polynomial $C_{m,\varepsilon}(u)$ of degree m such that*

$$L_{Q_{2m+\varepsilon}}(k) = (2k+1)^{\varepsilon} C_{m,\varepsilon}(k(k+1)).$$

More precisely,

$$C_{m,\varepsilon}(u) = \frac{1}{(2m+\varepsilon)!} \sum_{q=0}^{m} (-1)^{m-q} \binom{2m+\varepsilon+1}{m-q} \lambda_q^{\varepsilon} \prod_{a=0}^{m-1} \left(4\lambda_q^2 u + \lambda_q^2 - \lambda_a^2\right). \tag{2}$$

Proof. In (1), set $j = m - q$. Since

$$2m + \varepsilon + 1 - 2j = 2\lambda_q, \qquad 2m + \varepsilon - j = m + q + \varepsilon,$$

we get

$$L_{Q_{2m+\varepsilon}}(k) = \sum_{q=0}^{m} (-1)^{m-q} \binom{2m+\varepsilon+1}{m-q} \binom{2\lambda_q k + m + q + \varepsilon}{2m+\varepsilon}.$$

Hence it is enough to show that

$$\binom{2\lambda_q k + m + q + \varepsilon}{2m+\varepsilon} = \frac{\lambda_q^{\varepsilon}(2k+1)^{\varepsilon}}{(2m+\varepsilon)!} \prod_{a=0}^{m-1} \left(4\lambda_q^2 u + \lambda_q^2 - \lambda_a^2\right), \tag{3}$$

where $u = k(k+1)$. Let $X = \lambda_q(2k+1)$. Then

$$2\lambda_q k + m + q + \varepsilon = X + m + \frac{\varepsilon - 1}{2}.$$

If $\varepsilon = 0$, then the left-hand side of (3) can be written as

$$\binom{X + m - \frac{1}{2}}{2m} = \frac{(X + m - \frac{1}{2}) \cdots (X - m + \frac{1}{2})}{(2m)!} = \frac{\prod_{a=0}^{m-1} (X + \lambda_a)(X - \lambda_a)}{(2m)!}.$$

If $\varepsilon = 1$, then the left-hand side of (3) can be written as

$$\binom{X+m}{2m+1} = \frac{(X+m) \cdots X \cdots (X-m)}{(2m+1)!} = \frac{X \prod_{a=0}^{m-1} (X + \lambda_a)(X - \lambda_a)}{(2m+1)!}.$$

Since

$$X^2 - \lambda_a^2 = \lambda_q^2(4k^2+4k+1) - \lambda_a^2 = 4\lambda_q^2 u + \lambda_q^2 - \lambda_a^2,$$

substitution gives (3) in both cases. □

3. Divided differences

We collect the standard facts about divided differences that are used later. We refer to [7, 8] for general background. In particular, the mean-value theorem, the node-removal identity, the divided differences of monomials, and the reciprocal formula used below appear as Theorems 6, 7, 12, and 13 of [8].

Since we also use $[x^n]F(x)$ for coefficient extraction, we denote divided differences by the symbol Δ. The points at which a divided difference is taken are called its *nodes*. For a function f defined at pairwise distinct nodes $y_0, \dots, y_r$, set

$$\Delta[y_0, \dots, y_r]f := \sum_{a=0}^{r} \frac{f(y_a)}{\prod_{b\neq a}(y_a - y_b)}. \tag{4}$$

This formula shows that divided differences are symmetric in their nodes and linear in the function f. They also satisfy the recursive formula

$$\Delta[y_0, \dots, y_r]f = \frac{\Delta[y_1, \dots, y_r]f - \Delta[y_0, \dots, y_{r-1}]f}{y_r - y_0}. \tag{5}$$

This is also a consequence of [8, Theorem 7].

By linearity and [8, Theorem 12(i)], if $P(t)$ is a polynomial of degree at most $r-1$, then

$$\Delta[y_0, \dots, y_r]P = 0. \tag{6}$$

Lemma 3.1. *Let $r \geq 1$, and let $y_0, \dots, y_r$ be pairwise distinct positive real numbers. Then*

$$\Delta[y_0, \dots, y_r]t^{r-1/2} > 0.$$

Proof. By the mean-value theorem for divided differences [8, Theorem 6], there exists a point ξ in the positive interval spanned by the nodes such that

$$\Delta[y_0, \dots, y_r]t^{r-1/2} = \frac{1}{r!}\,\frac{d^r}{dt^r}t^{r-1/2}\bigg|_{t=\xi}.$$

For $t > 0$,

$$\frac{d^r}{dt^r}t^{r-1/2} = \left(r - \frac{1}{2}\right)\left(r - \frac{3}{2}\right)\cdots\frac{1}{2}\,t^{-1/2} > 0.$$

Thus we have $\Delta[y_0, \dots, y_r]t^{r-1/2} > 0$. □

The following node-removal identity is obtained by repeated application of [8, Theorem 7].

Lemma 3.2. *Let $y_0, y_1, \dots, y_s$ be pairwise distinct, let $L \subseteq \{1, \dots, s\}$, and write*

$$\{1, \dots, s\} \setminus L = \{\ell_1 < \cdots < \ell_q\}.$$

For every function h defined at $y_0, y_1, \dots, y_s$,

$$\Delta[y_0, y_1, \dots, y_s]\left(\prod_{b\in L}(t - y_b)h(t)\right) = \Delta[y_0, y_{\ell_1}, \dots, y_{\ell_q}]h.$$

Proof. Using (4), the summands corresponding to nodes y_b with $b \in L$ vanish. In each remaining summand, the factors corresponding to the removed nodes cancel between the numerator and denominator. The resulting sum is precisely the divided difference on the right-hand side. □

For example,

$$\Delta[y_0, y_1, y_2]\big((t - y_1)h(t)\big) = \Delta[y_0, y_2]h.$$

We also record the divided difference of a reciprocal linear factor.

Lemma 3.3. *Let $y_0, \dots, y_r$ be pairwise distinct and suppose that $y_a + u \neq 0$ for $0 \leq a \leq r$. Then*

$$\Delta[y_0, \dots, y_r]\frac{1}{t+u} = \frac{(-1)^r}{\prod_{a=0}^{r}(y_a + u)}. \tag{7}$$

Proof. Set $x_a = y_a + u$. Formula (4) immediately gives translation invariance in the form

$$\Delta[y_0, \dots, y_r] f(t+u) = \Delta[x_0, \dots, x_r] f(t).$$

Applying [8, Theorem 13] to $f(t) = 1/t$, we obtain

$$\Delta[y_0, \dots, y_r] \frac{1}{t+u} = \Delta[x_0, \dots, x_r] \frac{1}{t} = \frac{(-1)^r}{x_0 x_1 \cdots x_r} = \frac{(-1)^r}{\prod_{a=0}^r (y_a + u)}.$$

□

The following form will be used in the proof of Lemma 5.1.

Corollary 3.4. *Let $y_0, \dots, y_r$ be pairwise distinct and suppose that $y_a + u \neq 0$ for all a. Then*

$$\Delta[y_0, \dots, y_r] \frac{\prod_{b=1}^r (t + y_b)}{t+u} = \frac{\prod_{b=1}^r (u - y_b)}{\prod_{a=0}^r (u + y_a)}.$$

Proof. For $r = 0$, the identity is simply evaluation at y_0. Assume $r \geq 1$ and set

$$q(t) := \prod_{b=1}^r (t + y_b).$$

Since $q(t) - q(-u)$ is divisible by $t + u$, there is a polynomial $Q_u(t)$ of degree at most $r - 1$ such that

$$q(t) = (t+u) Q_u(t) + q(-u).$$

Hence

$$\frac{q(t)}{t+u} = Q_u(t) + \frac{q(-u)}{t+u}.$$

By (6), the r-th divided difference of Q_u vanishes. Therefore Lemma 3.3 gives

$$\begin{aligned} \Delta[y_0, \dots, y_r] \frac{q(t)}{t+u} &= q(-u) \Delta[y_0, \dots, y_r] \frac{1}{t+u} \\ &= \left(\prod_{b=1}^r (y_b - u) \right) \frac{(-1)^r}{\prod_{a=0}^r (y_a + u)} \\ &= \frac{\prod_{b=1}^r (u - y_b)}{\prod_{a=0}^r (u + y_a)}. \end{aligned}$$

□

4. Local decompositions

We use the rising factorial

$$(a)_j = a(a+1) \cdots (a+j-1), \qquad (a)_0 = 1.$$

4.1. **A local expansion.** Let $S_m := \{0, 1, \dots, m-1\}$. Continue to use the notation

$$\alpha = \frac{1+\varepsilon}{2}, \qquad \lambda_q = q + \alpha.$$

For $J \subseteq S_m$, write $r = |J|$ and define

$$\omega_{m,\varepsilon}(J) := 4^r (m + \alpha) \Delta[\lambda_m^2, (\lambda_j^2)_{j \in J}] t^{r - 1/2}. \tag{8}$$

Proposition 4.1. *For $\varepsilon \in \{0, 1\}$,*

$$C_{m,\varepsilon}(u) = \sum_{J \subseteq S_m} \omega_{m,\varepsilon}(J) u^{|J|}.$$

Moreover, $\omega_{m,\varepsilon}(\emptyset) = 1$ and

$$\omega_{m,\varepsilon}(J) > 0 \qquad (J \subseteq S_m).$$

Proof. Since

$$\prod_{a=0}^{m-1}\left(4\lambda_q^2 u+\lambda_q^2-\lambda_a^2\right)=\sum_{J\subseteq S_m}\left(4^{|J|}\lambda_q^{2|J|}u^{|J|}\prod_{a\in S_m\setminus J}(\lambda_q^2-\lambda_a^2)\right),$$

we have

$$C_{m,\varepsilon}(u)=\sum_{J\subseteq S_m}\nu_{m,\varepsilon}(J)u^{|J|},$$

where

$$\nu_{m,\varepsilon}(J)=\frac{4^{|J|}}{(2m+\varepsilon)!}\sum_{q=0}^{m}(-1)^{m-q}\binom{2m+\varepsilon+1}{m-q}\lambda_q^{2|J|+\varepsilon}\prod_{a\in S_m\setminus J}(\lambda_q^2-\lambda_a^2).$$

If $q\in S_m\setminus J$, the product contains $\lambda_q^2-\lambda_q^2=0$. Thus we have

$$\nu_{m,\varepsilon}(J)=\frac{4^{|J|}}{(2m+\varepsilon)!}\sum_{q\in J\cup\{m\}}(-1)^{m-q}\binom{2m+\varepsilon+1}{m-q}\lambda_q^{2|J|+\varepsilon}\prod_{a\in S_m\setminus J}(\lambda_q^2-\lambda_a^2).$$

For $0\le q\le m$,

$$\begin{aligned}\prod_{\substack{0\le a\le m\\ a\neq q}}(\lambda_q^2-\lambda_a^2)&=\prod_{\substack{0\le a\le m\\ a\neq q}}(q-a)(q+a+2\alpha)\\ &=(-1)^{m-q}q!(m-q)!\frac{(q+2\alpha)_{m+1}}{2\lambda_q}\end{aligned}\tag{9}$$

since

$$\prod_{\substack{0\le a\le m\\ a\neq q}}(q-a)=q(q-1)\cdots 1\cdot(-1)\cdots(-m+q)=(-1)^{m-q}q!(m-q)!,$$

$$\prod_{\substack{0\le a\le m\\ a\neq q}}(q+a+2\alpha)=\frac{(q+2\alpha)_{m+1}}{2\lambda_q}.$$

For $q\in J\cup\{m\}$,

$$\prod_{a\in S_m\setminus J}(\lambda_q^2-\lambda_a^2)=\frac{\displaystyle\prod_{\substack{0\le a\le m\\ a\neq q}}(\lambda_q^2-\lambda_a^2)}{\displaystyle\prod_{\substack{c\in J\cup\{m\}\\ c\neq q}}(\lambda_q^2-\lambda_c^2)}.$$

Using (9), together with

$$\begin{aligned}(q+2\alpha)_{m+1}&=(q+\varepsilon+1)_{m+1}=\frac{(m+q+\varepsilon+1)!}{(q+\varepsilon)!},\\ \frac{q!}{(q+\varepsilon)!}&=\begin{cases}1 &= \lambda_q^{-\varepsilon} \quad \text{if } \varepsilon=0,\\ \frac{1}{q+1} &= \lambda_q^{-\varepsilon} \quad \text{if } \varepsilon=1,\end{cases}\\ \frac{(2m+\varepsilon+1)!}{(2m+\varepsilon)!}&=2m+\varepsilon+1=2(m+\alpha),\end{aligned}$$

we obtain

$$\nu_{m,\varepsilon}(J)=4^{|J|}(m+\alpha)\sum_{q\in J\cup\{m\}}\frac{\lambda_q^{2|J|-1}}{\displaystyle\prod_{\substack{c\in J\cup\{m\}\\ c\neq q}}(\lambda_q^2-\lambda_c^2)}.$$

The last sum is precisely the divided difference

$$\Delta[\lambda_m^2,(\lambda_j^2)_{j\in J}]t^{|J|-1/2}.$$

Thus

$$\nu_{m,\varepsilon}(J)=\omega_{m,\varepsilon}(J),$$

which proves the expansion.

For $J = \emptyset$,

$$\omega_{m,\varepsilon}(\emptyset) = (m+\alpha)\Delta[(m+\alpha)^2]t^{-1/2} = 1.$$

For $J \neq \emptyset$, positivity follows from Lemma 3.1. Hence every weight is positive. □

4.2. **Magic coefficients and their local decomposition.** The positivity in Proposition 4.1 does not by itself imply magic positivity, since the change to the magic basis introduces alternating signs. We therefore decompose the magic coefficients into local contributions, working with arbitrary distinct positive nodes with one distinguished maximum.

Let $m \geq 1$, let $S_m = \{0, \dots, m-1\}$, and let

$$X, x_0, \dots, x_{m-1}$$

be pairwise distinct positive real numbers such that

$$X > x_j \qquad (j \in S_m).$$

Write $\mathbf{x} = (X, x_0, \dots, x_{m-1})$. For $J \subseteq S_m$, define

$$\omega_{\mathbf{x}}(J) := 4^{|J|} X \Delta[X^2, (x_j^2)_{j\in J}] t^{|J|-1/2}, \qquad \Omega_{\mathbf{x}}(u) := \sum_{J\subseteq S_m} \omega_{\mathbf{x}}(J) u^{|J|}. \tag{10}$$

The magic transform of the polynomial $\Omega_{\mathbf{x}}(k(k+1))$ is

$$\mathcal{M}_{\mathbf{x}}(y) := (1-y)^{2m} \Omega_{\mathbf{x}}\left(\frac{y}{(1-y)^2}\right).$$

By (10),

$$\mathcal{M}_{\mathbf{x}}(y) = \sum_{J\subseteq S_m} \omega_{\mathbf{x}}(J) y^{|J|} (1-y)^{2m-2|J|}. \tag{11}$$

Write

$$\mathcal{M}_{\mathbf{x}}(y) = \sum_{s=0}^{2m} d_s y^s.$$

For $I \subseteq S_m$ with $|I| = s \leq m$, define

$$\tau_{\mathbf{x}}(I) := \sum_{J\subseteq I} (-1)^{s-|J|} \frac{\binom{2m-2|J|}{s-|J|}}{\binom{m-|J|}{s-|J|}} \omega_{\mathbf{x}}(J). \tag{12}$$

Proposition 4.2. *For $0 \leq s \leq m$,*

$$d_s = \sum_{\substack{I\subseteq S_m \\ |I|=s}} \tau_{\mathbf{x}}(I).$$

Moreover, $d_s = d_{2m-s}$.

Proof. The coefficient of y^s in the summand of (11) indexed by J is

$$(-1)^{s-|J|} \binom{2m-2|J|}{s-|J|} \omega_{\mathbf{x}}(J).$$

A fixed J with $|J| \leq s$ is contained in exactly

$$\binom{m-|J|}{s-|J|}$$

subsets $I \subseteq S_m$ of cardinality s. Summing (12) over all such I therefore gives d_s.

For the symmetry, each summand in (11) satisfies

$$y^{2m}\left((1/y)^{|J|}(1-1/y)^{2m-2|J|}\right) = y^{|J|}(1-y)^{2m-2|J|},$$

since $2m-2|J|$ is even. Hence $y^{2m}\mathcal{M}_{\mathbf{x}}(1/y) = \mathcal{M}_{\mathbf{x}}(y)$. □

4.3. **A divided-difference formula for the local coefficients.** For $0 \le q \le s$, let

$$e_q(z_1,\dots,z_s) = \sum_{\substack{A\subseteq\{1,\dots,s\}\\ |A|=q}} \prod_{b\in A} z_b, \qquad e_0 = 1,$$

be the elementary symmetric polynomial. For the terminating hypergeometric identity below, see [1, Chapter 2]. For $0 \le j \le s \le m$, define

$$\beta_{m,s,j} := \sum_{\ell=0}^{j} (-1)^{j-\ell} \binom{j}{\ell} 4^{\ell} \frac{\binom{2m-2\ell}{s-\ell}}{\binom{m-\ell}{s-\ell}}.$$

Lemma 4.3. *Let $a = m - s + \frac{1}{2}$. Then*

$$\beta_{m,s,j} = \frac{\binom{2m}{s}}{\binom{m}{s}} \frac{(a)_j (a)_{s-j}}{(a)_s}.$$

In particular,

$$\beta_{m,s,j} > 0 \qquad \text{and} \qquad \beta_{m,s,j} = \beta_{m,s,s-j}.$$

Proof. Set

$$C_\ell = \frac{\binom{2m-2\ell}{s-\ell}}{\binom{m-\ell}{s-\ell}}, \qquad C_0 = \frac{\binom{2m}{s}}{\binom{m}{s}}.$$

A direct comparison of successive ratios gives

$$\frac{4^\ell C_\ell}{C_0} = \frac{(1-2a-s)_\ell}{(1-a-s)_\ell}.$$

Therefore

$$\frac{\beta_{m,s,j}}{C_0} = (-1)^j \sum_{\ell=0}^{j} \frac{(-j)_\ell (1-2a-s)_\ell}{(1-a-s)_\ell \, \ell!} = (-1)^j \frac{(a)_j}{(1-a-s)_j},$$

where the second equality is the terminating Chu–Vandermonde identity. Finally,

$$(1-a-s)_j = (-1)^j (a+s-j)_j = (-1)^j \frac{(a)_s}{(a)_{s-j}},$$

which proves the formula. □

Proposition 4.4. *Let $I = \{i_1 < \cdots < i_s\} \subseteq S_m$, and put*

$$z_0 = X^2, \qquad z_b = x_{i_b}^2 \quad (1 \le b \le s).$$

Then

$$\tau_{\mathbf{x}}(I) = X C_{m,s} \Delta[z_0,\dots,z_s] \frac{\displaystyle\sum_{j=0}^{s} (a)_j (a)_{s-j} e_{s-j}(z_1,\dots,z_s) t^j}{\sqrt{t}}, \tag{13}$$

where

$$a = m - s + \frac{1}{2}, \qquad C_{m,s} = \frac{\binom{2m}{s}}{\binom{m}{s}(a)_s} > 0.$$

Proof. For $J \subseteq I$, let

$$\sigma(J) = \{b \in \{1,\dots,s\} : i_b \in J\}.$$

If $|J| = r$, then by Lemma 3.2,

$$\begin{aligned} \omega_{\mathbf{x}}(J) &= 4^r X \Delta[z_0, (z_b)_{b\in\sigma(J)}] t^{r-1/2} \\ &= 4^r X \Delta[z_0, z_1, \dots, z_s] \left(t^{r-1/2} \prod_{b\notin\sigma(J)} (t - z_b) \right). \end{aligned}$$

Substituting this into (12), grouping by $r = |J|$, and using

$$\sum_{\substack{J\subseteq I\\ |J|=r}} \prod_{b\notin\sigma(J)} (t - z_b) = e_{s-r}(t - z_1, \dots, t - z_s),$$

we obtain

$$\tau_{\mathbf{x}}(I) = X\Delta[z_0,\dots,z_s]\frac{\widetilde{P}(t)}{\sqrt{t}},$$

where

$$\widetilde{P}(t) = \sum_{r=0}^{s}(-1)^{s-r}\frac{\binom{2m-2r}{s-r}}{\binom{m-r}{s-r}}(4t)^r e_{s-r}(t-z_1,\dots,t-z_s).$$

The elementary identity

$$e_{s-r}(t-z_1,\dots,t-z_s) = \sum_{j=r}^{s}(-1)^{s-j}\binom{j}{r}e_{s-j}(z_1,\dots,z_s)t^{j-r}$$

shows that the coefficient of

$$e_{s-j}(z_1,\dots,z_s)t^j$$

in $\widetilde{P}(t)$ is $\beta_{m,s,j}$. The result now follows from Lemma 4.3. □

5. Positivity of the local coefficients

5.1. **A positivity lemma for divided differences.** The following inequality is the key positivity statement used below.

Lemma 5.1. *Let $r \geq 0$ and $0 < y_r < \cdots < y_1 < y_0$. Then*

$$\Delta[y_0,\dots,y_r]\frac{\prod_{b=1}^{r}(t+y_b)}{\sqrt{t}} > 0.$$

Proof. The case $r = 0$ is immediate. Assume $r \geq 1$ and set

$$q(t) := \prod_{b=1}^{r}(t+y_b), \qquad K_r := \Delta[y_0,\dots,y_r]\frac{q(t)}{\sqrt{t}}.$$

We use the elementary formula

$$\frac{1}{\sqrt{t}} = \frac{2}{\pi}\int_0^\infty \frac{dv}{t+v^2}, \qquad t > 0,$$

which follows from Euler's beta integral; see, for example, [1, Chapter 1]. Since the integral converges absolutely at every positive node and a divided difference at fixed distinct nodes is a finite linear combination of node evaluations, we may interchange the finite sum and the integral. Hence

$$K_r = \frac{2}{\pi}\int_0^\infty \Delta[y_0,\dots,y_r]\frac{q(t)}{t+v^2}\,dv.$$

By Corollary 3.4,

$$\Delta[y_0,\dots,y_r]\frac{q(t)}{t+v^2} = \frac{\prod_{b=1}^{r}(v^2-y_b)}{\prod_{a=0}^{r}(v^2+y_a)}.$$

Writing

$$y_a = x_a^2, \qquad x_0 > x_1 > \cdots > x_r > 0,$$

we obtain

$$K_r = \frac{2}{\pi}\int_0^\infty f(v)\,dv,$$

where

$$f(v) := \frac{1}{v^2+x_0^2}\prod_{b=1}^{r}\frac{v^2-x_b^2}{v^2+x_b^2}.$$

By substituting $v = x_0^2/w$, we have

$$\int_0^{x_0} f(v)\,dv = \int_{x_0}^\infty f\left(\frac{x_0^2}{w}\right)\frac{x_0^2}{w^2}dw.$$

Renaming the integration variable w as v, we obtain

$$\begin{aligned}\int_0^\infty f(v)\,dv &= \int_{x_0}^\infty \left(f(v) + \frac{x_0^2}{v^2} f\left(\frac{x_0^2}{v}\right)\right) dv \\ &= \int_{x_0}^\infty \frac{1}{v^2+x_0^2}\left(\prod_{b=1}^r \frac{v^2-x_b^2}{v^2+x_b^2} + \prod_{b=1}^r \frac{x_0^4 - x_b^2 v^2}{x_0^4 + x_b^2 v^2}\right) dv \\ &= \int_{x_0}^\infty \frac{1}{v^2+x_0^2}\left(\prod_{b=1}^r A_b + \prod_{b=1}^r B_b\right) dv,\end{aligned} \tag{14}$$

where

$$T := \frac{v^2}{x_0^2} \geq 1, \qquad s_b := \frac{x_b^2}{x_0^2} \in (0,1), \qquad A_b := \frac{T - s_b}{T + s_b}, \qquad B_b := \frac{1 - s_b T}{1 + s_b T}$$

for $v \geq x_0$ and $b = 1, \ldots, r$. Since $T \geq 1$ and $0 < s_b < 1$, we have

$$2s_b(T^2 - 1) \geq 0, \qquad 2T(1 - s_b^2) \geq 0.$$

Thus it follows that

$$B_b \leq A_b, \qquad -A_b \leq B_b,$$

and hence

$$|B_b| \leq A_b.$$

Therefore $|\prod_{b=1}^r B_b| \leq \prod_{b=1}^r A_b$, and so

$$\prod_{b=1}^r A_b + \prod_{b=1}^r B_b \geq 0.$$

Thus the integrand in (14) is nonnegative for all $v \geq x_0$. At $v = x_0$, equivalently $T = 1$, we have

$$A_b = B_b = \frac{1 - s_b}{1 + s_b} > 0,$$

and hence the integrand in (14) is strictly positive there. By continuity it is positive on a neighborhood of x_0. Hence the integral in (14) is strictly positive. Consequently,

$$K_r = \frac{2}{\pi}\int_0^\infty f(v)\,dv > 0,$$

as desired. □

5.2. **Finite symmetrization and local positivity.** To apply Lemma 5.1 to the expression in Proposition 4.4, we rewrite its numerator as a positive linear combination of products involving factors $t - z_b$ and $t + z_b$. The factors $t - z_b$ can then be eliminated by Lemma 3.2. We obtain the required identity using binary Krawtchouk polynomials. For a fixed $s \geq 0$ and $0 \leq q, j \leq s$, define

$$\kappa_q^{(s)}(j) := [z^q](1 - z)^j (1 + z)^{s-j}. \tag{15}$$

Equivalently,

$$\kappa_q^{(s)}(j) = \sum_{h=0}^{q} (-1)^h \binom{j}{h}\binom{s-j}{q-h}. \tag{16}$$

These are the binary Krawtchouk polynomials in the normalization used by Feinsilver and Kocik; see [11]. Equivalently, they are the entries of the Krawtchouk matrix of order s.

The following is the standard orthogonality relation for these polynomials, written in the dual form needed below. We include the short derivation for completeness. Here $\delta_{i,j}$ denotes the Kronecker delta.

Lemma 5.2. *For $0 \leq i, j \leq s$,*

$$\frac{1}{2^s}\sum_{q=0}^{s} \frac{\kappa_q^{(s)}(i)\kappa_q^{(s)}(j)}{\binom{s}{q}} = \frac{\delta_{i,j}}{\binom{s}{i}}. \tag{17}$$

Proof. By the binomial theorem,

$$\begin{aligned}&\sum_{j=0}^{s}\binom{s}{j}(1-z)^j(1+z)^{s-j}(1-w)^j(1+w)^{s-j}\\&=((1-z)(1-w)+(1+z)(1+w))^s\\&=2^s(1+zw)^s.\end{aligned}$$

Comparing coefficients of $z^p w^q$ and using (15), we obtain

$$\sum_{j=0}^{s}\binom{s}{j}\kappa_p^{(s)}(j)\kappa_q^{(s)}(j)=2^s\binom{s}{q}\delta_{p,q}.$$

Let

$$K=(\kappa_q^{(s)}(j))_{0\le j,q\le s},\qquad B=\operatorname{diag}\left(\binom{s}{0},\dots,\binom{s}{s}\right).$$

The preceding relation is

$$K^{\mathsf{T}}BK=2^sB.$$

In particular, K is invertible. Taking inverses gives

$$K^{-1}B^{-1}(K^{\mathsf{T}})^{-1}=2^{-s}B^{-1}.$$

Multiplying on the left by K and on the right by K^{T}, we obtain

$$KB^{-1}K^{\mathsf{T}}=2^sB^{-1},$$

which is precisely (17). □

We now compute the Krawtchouk expansion of the specific symmetric weights that occur in the local positivity argument. This is the specialization needed for the finite symmetrization identity.

Lemma 5.3. *Let $a>0$ and $s\ge 0$. For $0\le j\le s$, put*

$$W_j:=\frac{(a)_j(a)_{s-j}}{(2a)_s}.$$

Then

$$W_j=\frac{1}{2^s}\sum_{\ell=0}^{\lfloor s/2\rfloor}\frac{(1/2)_\ell}{(a+1/2)_\ell}\kappa_{2\ell}^{(s)}(j). \tag{18}$$

Consequently, for indeterminates $t,y_1,\dots,y_s$,

$$\begin{aligned}&\sum_{j=0}^{s}(a)_j(a)_{s-j}e_{s-j}(y_1,\dots,y_s)t^j\\&\quad=\frac{(2a)_s}{2^s}\sum_{\substack{L\subseteq\{1,\dots,s\}\\|L|\ \text{even}}}\frac{(1/2)_{|L|/2}}{(a+1/2)_{|L|/2}}\prod_{b\in L}(t-y_b)\prod_{b\notin L}(t+y_b).\end{aligned} \tag{19}$$

Proof. We first prove (18). Consider the (unnormalized) Krawtchouk transform

$$\widehat{W}_q:=\sum_{j=0}^{s}\binom{s}{j}W_j\kappa_q^{(s)}(j).$$

By (15),

$$\sum_{q=0}^{s}\widehat{W}_qz^q=\sum_{j=0}^{s}\binom{s}{j}\frac{(a)_j(a)_{s-j}}{(2a)_s}(1-z)^j(1+z)^{s-j}. \tag{20}$$

Using the formal power series

$$(1-XT)^{-a}=\sum_{n\ge0}\frac{(a)_n}{n!}X^nT^n,$$

we obtain, for arbitrary X, Y,

$$\sum_{j=0}^{s}\binom{s}{j}(a)_j(a)_{s-j}X^jY^{s-j} = s!\sum_{j=0}^{s}\frac{(a)_j}{j!}X^j\frac{(a)_{s-j}}{(s-j)!}Y^{s-j} \tag{21}$$
$$= s![T^s](1-XT)^{-a}(1-YT)^{-a}.$$

Substituting $X = 1-z$ and $Y = 1+z$ into (21), expression (20) becomes

$$\frac{s!}{(2a)_s}[T^s]\left((1-T)^2 - z^2T^2\right)^{-a}.$$

Applying

$$(1-X)^{-a} = \sum_{\ell\geq 0}\frac{(a)_\ell}{\ell!}X^\ell,$$

for $X = \frac{z^2T^2}{(1-T)^2}$, we have

$$\left((1-T)^2 - z^2T^2\right)^{-a} = (1-T)^{-2a}\left(1 - \frac{z^2T^2}{(1-T)^2}\right)^{-a}$$
$$= \sum_{\ell\geq 0}\frac{(a)_\ell}{\ell!}z^{2\ell}T^{2\ell}(1-T)^{-2a-2\ell}.$$

Since only even powers of z occur in the above expansion,

$$\widehat{W}_{2\ell+1} = 0.$$

In addition, for $0 \leq 2\ell \leq s$, since

$$[T^s]T^{2\ell}(1-T)^{-2a-2\ell} = [T^{s-2\ell}]\sum_{n\geq 0}\frac{(2a+2\ell)_n}{n!}T^n = \frac{(2a+2\ell)_{s-2\ell}}{(s-2\ell)!}$$

and since

$$(2a)_s = (2a)_{2\ell}(2a+2\ell)_{s-2\ell},$$

we have

$$\widehat{W}_{2\ell} = \frac{s!}{(2a)_s}\frac{(a)_\ell}{\ell!}\frac{(2a+2\ell)_{s-2\ell}}{(s-2\ell)!} = \frac{s!}{(s-2\ell)!}\frac{(a)_\ell}{\ell!\,(2a)_{2\ell}} = \binom{s}{2\ell}\frac{(2\ell)!}{\ell!}\frac{(a)_\ell}{(2a)_{2\ell}}.$$

Using the duplication identities

$$(2a)_{2\ell} = 2^{2\ell}(a)_\ell\left(a+\frac{1}{2}\right)_\ell, \qquad (2\ell)! = 2^{2\ell}\ell!\left(\frac{1}{2}\right)_\ell,$$

it follows that

$$\frac{(2\ell)!}{\ell!}\frac{(a)_\ell}{(2a)_{2\ell}} = \frac{2^{2\ell}(\frac{1}{2})_\ell(a)_\ell}{2^{2\ell}(a)_\ell(a+\frac{1}{2})_\ell} = \frac{(\frac{1}{2})_\ell}{(a+\frac{1}{2})_\ell}.$$

Hence

$$\widehat{W}_{2\ell} = \binom{s}{2\ell}\frac{(\frac{1}{2})_\ell}{(a+\frac{1}{2})_\ell}. \tag{22}$$

Then

$$\sum_{q=0}^{s}\frac{\kappa_q^{(s)}(i)}{2^s\binom{s}{q}}\widehat{W}_q = \sum_{q=0}^{s}\frac{\kappa_q^{(s)}(i)}{2^s\binom{s}{q}}\sum_{j=0}^{s}\binom{s}{j}W_j\kappa_q^{(s)}(j)$$
$$= \sum_{j=0}^{s}\binom{s}{j}W_j\sum_{q=0}^{s}\frac{\kappa_q^{(s)}(i)\kappa_q^{(s)}(j)}{2^s\binom{s}{q}}.$$

By Lemma 5.2, this equals

$$\sum_{j=0}^{s}\binom{s}{j}W_j\frac{\delta_{i,j}}{\binom{s}{i}} = W_i.$$

Substituting $\widehat{W}_{2\ell+1} = 0$ and (22), we have

$$W_i = \sum_{q=0}^{s}\frac{\kappa_q^{(s)}(i)}{2^s\binom{s}{q}}\widehat{W}_q = \sum_{\ell=0}^{\lfloor s/2\rfloor}\frac{\kappa_{2\ell}^{(s)}(i)}{2^s\binom{s}{2\ell}}\widehat{W}_{2\ell} = \sum_{\ell=0}^{\lfloor s/2\rfloor}\frac{\kappa_{2\ell}^{(s)}(i)}{2^s}\frac{(\frac{1}{2})_\ell}{(a+\frac{1}{2})_\ell}$$

which yields (18).

We next prove (19). For $L \subseteq \{1, \dots, s\}$, set

$$F_L(t, y) := \prod_{b \in L} (t - y_b) \prod_{b \notin L} (t + y_b).$$

Fix j and ℓ, and consider

$$\sum_{|L|=2\ell} F_L(t, y).$$

This polynomial is symmetric in $y_1, \dots, y_s$, and it is multi-affine in these variables. Its coefficient of t^j is therefore a symmetric multi-affine homogeneous polynomial of degree $s - j$. Every monomial in this coefficient is a product of $s - j$ distinct variables, and symmetry forces all such monomials to have the same coefficient. Hence it is a scalar multiple of $e_{s-j}(y_1, \dots, y_s)$.

To obtain a monomial contributing to $e_{s-j}(y)t^j$, we must choose $s - j$ of the s factors to contribute a y-term and the remaining j factors to contribute t. Suppose that among the chosen $s - j$ factors, exactly h belong to L. Each such choice contributes the sign $(-1)^h$. The number of possibilities is

$$\binom{s-j}{h}\binom{j}{2\ell - h},$$

since h indices of L must lie among the $s - j$ chosen y-positions and the remaining $2\ell - h$ indices of L among the j chosen t-positions. Hence the coefficient of $e_{s-j}(y)t^j$ in $\sum_{|L|=2\ell} F_L(t, y)$ is

$$\sum_h (-1)^h \binom{s-j}{h}\binom{j}{2\ell - h}.$$

By the defining formula (16), this equals $\kappa^{(s)}_{2\ell}(s - j)$. Since

$$\kappa^{(s)}_q(s - j) = (-1)^q \kappa^{(s)}_q(j),$$

and $q = 2\ell$ is even, we obtain

$$\kappa^{(s)}_{2\ell}(s - j) = \kappa^{(s)}_{2\ell}(j).$$

Therefore the coefficient of $e_{s-j}(y)t^j$ on the right-hand side of (19) is

$$\frac{(2a)_s}{2^s} \sum_{\ell=0}^{\lfloor s/2 \rfloor} \frac{(1/2)_\ell}{(a + 1/2)_\ell} \kappa^{(s)}_{2\ell}(j).$$

By (18), this equals

$$(2a)_s W_j = (a)_j (a)_{s-j}.$$

Hence the coefficients of $e_{s-j}(y_1, \dots, y_s)t^j$ coincide on both sides of (19), which proves the identity. □

Theorem 5.4. *Let $m \geq 1$ and let $X, x_0, \dots, x_{m-1}$ be pairwise distinct positive real numbers satisfying $X > x_j$ for every j. Then $\tau_{\mathbf{x}}(I) > 0$ for every $I \subseteq S_m$. Consequently, every coefficient of*

$$\mathcal{M}_{\mathbf{x}}(y) = (1 - y)^{2m} \Omega_{\mathbf{x}}\left(\frac{y}{(1 - y)^2}\right)$$

is strictly positive.

Proof. Fix $I = \{i_1 < \cdots < i_s\} \subseteq S_m$ and use the notation of Proposition 4.4. In particular,

$$a = m - s + \frac{1}{2} > 0.$$

Applying Lemma 5.3 to the numerator in (13), and then using Lemma 3.2, gives

$$\tau_{\mathbf{x}}(I) = \frac{XC_{m,s}(2a)_s}{2^s} \sum_{\substack{L \subseteq \{1,\dots,s\} \\ |L| \text{ even}}} \frac{(1/2)_{|L|/2}}{(a + 1/2)_{|L|/2}} \Delta[z_0, (z_b)_{b \notin L}] \frac{\prod_{b \notin L}(t + z_b)}{\sqrt{t}}.$$

The distinguished node $z_0 = X^2$ is larger than all remaining positive nodes. By symmetry of divided differences, the latter may be arranged in decreasing order, and Lemma 5.1 shows that every divided difference in the sum is strictly positive. All other factors are positive as well. Hence $\tau_{\mathbf{x}}(I) > 0$.

Proposition 4.2 now gives $d_s > 0$ for $0 \leq s \leq m$, and the symmetry $d_s = d_{2m-s}$ gives the result for all coefficients. □

6. Applications

We first apply Theorem 5.4 to the nodes appearing in Proposition 4.1.

Theorem 6.1. *For every $d \geq 2$, all magic coefficients of $L_{Q_d}(k)$ are strictly positive.*

Proof. Write

$$d = 2m + \varepsilon, \qquad \varepsilon \in \{0, 1\}, \qquad \alpha = \frac{1+\varepsilon}{2}.$$

Apply Theorem 5.4 with

$$X = m + \alpha, \qquad x_j = j + \alpha \quad (0 \leq j \leq m-1).$$

Proposition 4.1 gives

$$C_{m,\varepsilon}(u) = \Omega_{\mathbf{x}}(u).$$

Hence

$$H_{m,\varepsilon}(y) := (1-y)^{2m} C_{m,\varepsilon}\left(\frac{y}{(1-y)^2}\right)$$

has strictly positive coefficients. By Proposition 2.5,

$$(1-y)^{2m+\varepsilon} L_{Q_{2m+\varepsilon}}\left(\frac{y}{1-y}\right) = (1+y)^{\varepsilon} H_{m,\varepsilon}(y).$$

Since ε is either 0 or 1, the right-hand side has strictly positive coefficients. These are precisely the magic coefficients of $L_{Q_d}(k)$. □

Since $V_{2m}^\vee = Q_{2m}$, the del Pezzo part of the classification is now settled. We next treat the pseudo-del Pezzo polytope.

Proposition 6.2. *For every $m \geq 1$,*

$$L_{\widetilde{V}_{2m}^\vee}(k) = \frac{(2k+1)^{2m} + L_{V_{2m}^\vee}(k)}{2}.$$

Proof. Put $d = 2m$. From the definition of $\widetilde{V}_d$,

$$\widetilde{V}_d^\vee = \{x \in [-1,1]^d : x_1 + \cdots + x_d \geq -1\}.$$

Thus $L_{\widetilde{V}_d^\vee}(k)$ counts vectors $z \in [-k,k]^d \cap \mathbb{Z}^d$ satisfying

$$z_1 + \cdots + z_d \geq -k.$$

Let $T = (2k+1)^d$ and let

$$B = \#\{z \in [-k,k]^d \cap \mathbb{Z}^d : -k \leq z_1 + \cdots + z_d \leq k\}.$$

The central band is $kV_d^\vee$, so $B = L_{V_d^\vee}(k)$. The involution $z \mapsto -z$ exchanges the regions where the coordinate sum is greater than k and less than $-k$. If their common cardinality is C, then

$$T = B + 2C, \qquad L_{\widetilde{V}_d^\vee}(k) = B + C,$$

which gives the formula. □

Theorem 6.3. *For every $m \geq 1$, all magic coefficients of $L_{V_{2m}^\vee}(k)$ and $L_{\widetilde{V}_{2m}^\vee}(k)$ are strictly positive.*

Proof. Let a_i and $\widetilde{a}_i$ denote the magic coefficients of $L_{V_{2m}^\vee}(k)$ and $L_{\widetilde{V}_{2m}^\vee}(k)$, respectively. By Theorem 6.1, we have $a_i > 0$ for $0 \leq i \leq 2m$. Since $(2k+1)^{2m} = (k+(k+1))^{2m}$, Proposition 6.2 gives

$$\widetilde{a}_i = \frac{\binom{2m}{i} + a_i}{2} > 0 \qquad (0 \leq i \leq 2m).$$

□

We now prove Theorem 1.2. By Proposition 2.2, every pseudo-symmetric simplicial reflexive polytope P is lattice equivalent to a free sum of a centrally symmetric reflexive crosspolytope and del Pezzo and pseudo-del Pezzo factors. Taking polar duals turns this decomposition into a Cartesian product. Proposition 2.4, Theorem 6.3, and Lemma 2.1 therefore show that $L_{P^\vee}(k)$ is magic positive.

Suppose now that P is smooth Fano. By Proposition 2.3, P is a free sum of copies of $[-1,1]$, del Pezzo polytopes, and pseudo-del Pezzo polytopes. The two magic coefficients of

$$L_{[-1,1]}(k) = 2k + 1 = k + (k+1)$$

are both equal to 1, while Theorem 6.3 gives strict positivity for the other factors. The strict version of Lemma 2.1(3) therefore shows that all magic coefficients of $L_{P^\vee}(k)$ are strictly positive. The lattice equivalence $\mathcal{P}_{C_n}^\vee \cong Q_{n-1}$ [17, Proposition 4.6 and its proof] gives Theorem 1.4 from Theorem 6.1. Since the polar duals appearing in both theorems are reflexive, Corollaries 1.3 and 1.5 follow from the implications of magic positivity recalled in the Introduction.

6.1. **Further questions.** Theorem 5.4 gives a local decomposition

$$d_s = \sum_{\substack{I \subseteq S_m \\ |I|=s}} \tau_{\mathbf{x}}(I), \qquad \tau_{\mathbf{x}}(I) > 0 \qquad (0 \le s \le m).$$

The remaining coefficients are determined by the symmetry $d_s = d_{2m-s}$. It would be interesting to find a direct combinatorial interpretation of these local coefficients in the arithmetic-node specializations arising from del Pezzo polytopes and cycles.

Question 6.4. *Do the local coefficients $\tau_{\mathbf{x}}(I)$ arising from $V_{2m}^\vee$ admit a natural weighted combinatorial interpretation related to its quadratic triangulations?*

Appendix A. A missing argument in a proof in [22]

In [22, Theorem 5.8], we stated that the h^*-polynomial of every pseudo-symmetric simplicial reflexive polytope is γ-positive. The proof there uses Nill's decomposition into a centrally symmetric reflexive crosspolytope, del Pezzo polytopes, and pseudo-del Pezzo polytopes. However, the centrally symmetric reflexive crosspolytope factor was inadvertently treated as a standard crosspolytope. We supply the missing argument here.

Proposition A.1. *Let R be a d-dimensional centrally symmetric reflexive crosspolytope, and let $C \subseteq \mathbb{F}_2^d$ be the even binary code associated with R as in Section 2.1. For $0 \le 2j \le d$, let*

$$A_{2j} = \#\{c \in C : \mathrm{wt}(c) = 2j\}.$$

Then

$$h^*(R;t) = \sum_{j=0}^{\lfloor d/2 \rfloor} 4^j A_{2j} t^j (1+t)^{d-2j}.$$

In particular, $h^*(R;t)$ is γ-positive.

Proof. For $x = (x_1, \dots, x_d) \in \mathbb{R}^d$, write $\|x\|_1 = |x_1| + \cdots + |x_d|$. Using the lattice realization $\mathbb{Z}^d \subseteq \Lambda \subseteq \frac{1}{2}\mathbb{Z}^d$ from Section 2.1, decompose Λ into cosets of $\mathbb{Z}^d$ indexed by C.

For an integral coordinate, $\sum_{n\in\mathbb{Z}} t^{|n|} = \frac{1+t}{1-t}$, whereas for a half-integral coordinate,

$$\sum_{n\in\mathbb{Z}} t^{|n+1/2|} = \frac{2t^{1/2}}{1-t}.$$

Hence a codeword of weight $2j$ contributes

$$\frac{4^j t^j (1+t)^{d-2j}}{(1-t)^d}$$

to $\sum_{x\in\Lambda} t^{\|x\|_1}$. Therefore

$$\sum_{x\in\Lambda} t^{\|x\|_1} = \frac{1}{(1-t)^d} \sum_{j=0}^{\lfloor d/2 \rfloor} 4^j A_{2j} t^j (1+t)^{d-2j}.$$

Since every sign vector belongs to $\Lambda^\vee$, for every $x \in \Lambda$ the value $\|x\|_1$ is an integer. Thus

$$\sum_{k\ge 0} L_R(k) t^k = \frac{1}{1-t} \sum_{x\in\Lambda} t^{\|x\|_1}.$$

Comparing this expression with the Ehrhart series

$$\sum_{k\ge 0} L_R(k) t^k = \frac{h^*(R;t)}{(1-t)^{d+1}}$$

gives the desired formula. □

Together with Nill's classification and the remaining argument in [22, Theorem 5.8], Proposition A.1 completes the proof of that theorem.

Declaration on the use of generative AI. OpenAI's ChatGPT was used for exploratory discussions of proof strategies, English-language editing, and LaTeX assistance. All mathematical arguments and references were verified by the authors.

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